

SUBJECTIVE TYPE

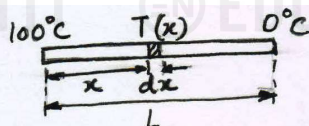
1. A uniform rod of cross section area 1 cm^2 and length 1 m is initially at a temperature of 0°C . It is then kept between 2 reservoirs of temperature 100°C and 0°C . Specific heat capacity is $10 \text{ J/kg}^\circ\text{C}$ and linear mass density is 2 kg/m . Find the



- temperature gradient along the rod in steady state
- total heat absorbed by the rod to reach steady state

Solution(a) In steady state, the temperature gradient along the rod is constant given by

$$\frac{dT}{dx} = \frac{0^\circ\text{C} - 100^\circ\text{C}}{1\text{m}} = -100^\circ\text{C/m}$$



(b) Consider a small element of length dx as shown in fig. Its mass is

$$dm = \lambda dx = 2dx$$

and its temperature is given by

$$\frac{100 - T(x)}{x} = \frac{100 - 0}{1} \Rightarrow T(x) = 100(1 - x)$$

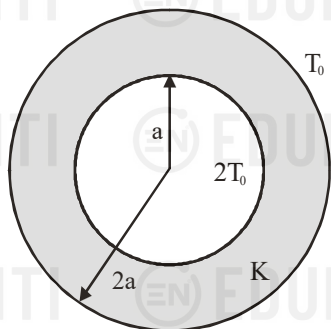
Heat absorbed by this element is

$$dQ = (dm)S(T - 0) = (2dx) \times 10 \times 100(1 - x)$$

The total heat absorbed by rod is

$$\Delta Q = 2000 \int_0^1 (1 - x) dx = 1000 \text{ J}$$

2. A hollow spherical shell of inner radius a and outer radius $2a$ is made of a uniform



material of constant thermal conductivity K . The temperature within the shell is maintained at $2T_0$ and outside the ball it is T_0 . Assuming steady state condition, find

- the rate at which heat flows out of the ball
- the temperature at $r = 3a/2$, where r is radial distance from the centre of shell.

Solution In the steady state, the net outward heat current H is constant and does not depend on the radial position.

(a) Consider a small elemental sphere of radius r and thickness dr . In steady state, the heat current through this elemental sphere is

$$H = -K(4\pi r^2) \frac{dT}{dr} \Rightarrow - \int_{2T_0}^T dT = \frac{H}{4\pi K} \int_a^r \frac{dr}{r^2}$$

where T is the temperature at distance r from the centre of the shell.

$$\Rightarrow 2T_0 - T = \frac{H}{4\pi K} \left(\frac{1}{a} - \frac{1}{r} \right)$$

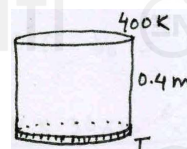
Putting $T = T_0$ at $r = 2a$, we get $H = 8\pi aKT_0$

$$(b) \text{ At } r = 3a/2, 2T_0 - T = \frac{8\pi aKT_0}{4\pi K} \left(\frac{1}{a} - \frac{1}{3a/2} \right) = \frac{2}{3}T_0$$

$$\therefore T = 4T_0/3$$

3. A cylindrical block of length 0.4 m and area of cross-section 0.04 m^2 is placed coaxially on a thin metal disc of mass 0.4 kg and of the same cross-section. The upper face of the cylinder is maintained at a constant temperature of 400 K and the initial temperature of the disc is 300 K . If the thermal conductivity of the material of the cylinder is 10 W/mK and the specific heat capacity of the material of the disc is 600 J/kg-K , how long will it take for the temperature of the disc to increase to 350 K ? Assume, for purposes of calculation, the heat capacity of the cylinder to be very low and the system to be thermally insulated except for the upper face of the cylinder.

Solution



Mass of disc, $m = 0.4 \text{ kg}$

Specific heat capacity of disc, $s = 600 \text{ J/kg-K}$

Length of cylinder, $l = 0.4 \text{ m}$

Area of cross section, $A = 0.04 \text{ m}^2$

Thermal conductivity of material of cylinder, $K = 10 \text{ W/mK}$

The rate at which the heat is absorbed by the disc is equal to the rate at which it is conducted through the cylinder.

Let at any instant t , the temperature of the disc be T . Then,

$$ms \frac{dT}{dt} = \frac{KA(400 - T)}{l}$$

$$\Rightarrow \frac{KA}{mls} \int_0^t dt = \int_{300}^{350} \frac{dT}{400 - T}$$

$$\Rightarrow \frac{KA}{mls} = \ln \left(\frac{400 - 300}{400 - 350} \right) = \ln 2$$

$$\therefore t = \frac{mls}{KA} \ln 2 = \frac{0.4 \times 0.4 \times 600 \times 0.693}{10 \times 0.04} = 166.3 \text{ s}$$

4. A hot body placed in air is cooled down according to Newton's law of cooling, the rate of decrease of temperature being k times the temperature difference between the body and the surrounding. Starting from $t = 0$, find the time in which the body will lose half the maximum heat it can lose.

Solution Let the initial temperature of the body be T_i and the temperature of the surrounding be T_0 . If T is the instantaneous temperature of the body at time t , then as per Newton's law of cooling,



LEVEL UP - JEE ADVANCED 2022

THERMAL PROPERTIES

EDUNITI

$$-\frac{dT}{dt} = k(T - T_0) \Rightarrow -\int_{T_i}^T \frac{dT}{T - T_0} = \int_0^t k dt$$

$$\Rightarrow t = \frac{1}{k} \ln \left(\frac{T_i - T_0}{T - T_0} \right) \quad \dots (1)$$

When the body loses the maximum heat, its temperature is T_0 and it has fallen by $T_i - T_0$. When the body loses half of the maximum heat, its temperature has fallen by $(T_i - T_0)/2$ and at this instant, its temperature is

$$T = T_i - \frac{T_i - T_0}{2} = \frac{T_i + T_0}{2}$$

On putting this in eqn (1), we obtain $t = \frac{\ln 2}{k}$

5. A cylindrical rod of heat capacity 120 J/K at room temperature 27°C is heated internally by a heater of power 240 W . The steady state temperature attained by the rod is 37°C . Now the heater is switched off. Assuming Newton's law of cooling to hold, find

- the initial rate of increase in temperature
- the steady state rate of emission of radiant heat.
- the rate of decrease in temperature when heater is switched off
- the rate of decrease in temperature of the cylinder when its temperature falls to 31°C
- the total amount of heat lost by the cylinder

Solution The heater provides heat at a constant rate $P = 240 \text{ W}$. Out of this, some heat is retained by the cylinder that increases its temperature and the balance is emitted as radiant heat. So,

$$P = S \frac{dT}{dt} + k(T - T_0) \Rightarrow P = 120 \frac{dT}{dt} + k(T - 27)$$

(a) Initially, $P = 240 \text{ W}$ and $T = 27^\circ\text{C}$.

$$\therefore \frac{dT}{dt} = \frac{240}{120} = 2^\circ\text{C/s}$$

(b) In steady state, $P = 240 \text{ W}$, $T = 37^\circ\text{C}$ and $dT/dt = 0$.

Therefore, the rate of emission of radiant heat is 240 W .

$$\text{Also, } k(37 - 27) = 240 \Rightarrow k = 24 \text{ W/}^\circ\text{C}$$

(c) When heater is switched off,

$$P = 0 \text{ and } T = 37^\circ\text{C}$$

$$\Rightarrow 0 = 120 \frac{dT}{dt} + 24(37 - 27) \Rightarrow \frac{dT}{dt} = -2^\circ\text{C/s}$$

The rate of decrease in temperature is 2°C/s

(d) When T falls to 31°C , $P = 0$ and $T = 31^\circ\text{C}$

$$\Rightarrow 0 = 120 \frac{dT}{dt} + 24(31 - 27) \Rightarrow \frac{dT}{dt} = -0.8^\circ\text{C/s}$$

The rate of decrease in temperature is 0.8°C/s

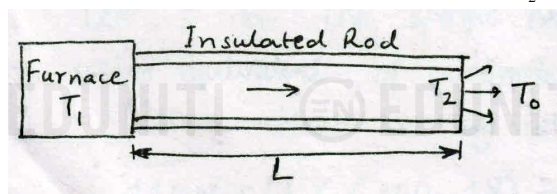
(e) The total heat lost by cylinder is

Heat capacity $\times$ Change in temperature

$$= 120 \times (37 - 27) = 1200 \text{ J}$$

6. One end of a rod of length L and cross-sectional area A is in a furnace at temperature T_1 . In steady state, the other end of the rod

is kept at a temperature T_2 . The thermal conductivity of the material of the rod is K and emissivity of the rod is e . It is given that $T_2 = T_0 + \Delta T$, where $\Delta T \ll T_0$ and T_0 is the temperature of the surroundings. If ΔT is proportional to $T_1 - T_2$, find the proportional constant. The rod is insulated and heat is lost only by radiation at the end where the temperature of the rod is T_2 .



Solution In steady state, the rate at which heat is conducted through the rod is equal to the rate at which it is radiated. So,

$$\frac{K A (T_1 - T_2)}{L} = e A \sigma (T_2^4 - T_0^4)$$

where,

$$T_2^4 - T_0^4 = (T_0 + \Delta T)^4 - T_0^4 = T_0^4 \left(1 + \frac{\Delta T}{T_0} \right)^4 - T_0^4$$

$$= T_0^4 \left(1 + 4 \frac{\Delta T}{T_0} \right) - T_0^4 = 4 T_0^3 \Delta T$$

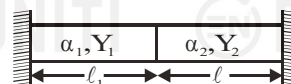
$$\Rightarrow \frac{K (T_1 - T_2)}{L} = e \sigma \times 4 T_0^3 \Delta T$$

$$\therefore \frac{\Delta T}{T_1 - T_2} = \frac{K}{e \sigma L T_0^3}$$

PARAGRAPH TYPE

Passage for Question Nons. 7 to 8

Two rods of equal cross sectional area are joined end to end as shown in figure. These are supported between two rigid vertical walls. Initially the rods are unstrained.



7. If temperature of system is increased by ΔT , then junction will not shift if

- $Y_1 \alpha_1 = Y_2 \alpha_2$
- $Y_1 \alpha_1 \ell_1 = Y_2 \alpha_2 \ell_2$
- $\alpha_1 = \alpha_2$
- $Y_2 \alpha_1 \ell_1 = Y_1 \alpha_2 \ell_2$

8. If temperature of system is increased by ΔT , then shifting in junction is given by (Assume that shifting is towards right)

$$(a) \frac{\ell_1 \ell_2 (Y_1 \alpha_2 - Y_2 \alpha_1) \Delta T}{Y_1 \ell_1 + Y_2 \ell_2} \quad (b) \frac{\ell_1 \ell_2 (Y_1 \alpha_1 - Y_2 \alpha_2) \Delta T}{Y_1 \ell_2 + Y_2 \ell_1}$$

$$(c) \frac{\ell_1 \ell_2 (Y_1 \alpha_1 - Y_2 \alpha_2) \Delta T}{Y_1 \ell_1 + Y_2 \ell_2} \quad (d) \frac{\ell_1 \ell_2 (Y_1 \alpha_2 - Y_2 \alpha_1) \Delta T}{Y_1 \ell_2 + Y_2 \ell_1}$$

Passage (Solution)

- 7(a) If the junction does not shift, the thermal stresses developed in the two rods are respectively

$$Y_1 \alpha_1 \Delta T \text{ and } Y_2 \alpha_2 \Delta T.$$

The compressive force developed in the two rods is always same. As the rods have equal cross sectional areas, the stress developed in them is also same. So,

$$Y_1 \alpha_1 \Delta T = Y_2 \alpha_2 \Delta T \quad \therefore Y_1 \alpha_1 = Y_2 \alpha_2$$

- 8(b) If the rods were free to expand, the length of rods at increased temperature would have been respectively

$$\ell_1 + \alpha_1 \ell_1 \Delta T \text{ and } \ell_2 + \alpha_2 \ell_2 \Delta T.$$

If the shifting of junction is $\Delta \ell$ towards right, the length of rods in stressed condition are respectively

$$\ell_1 + \Delta \ell \text{ and } \ell_2 - \Delta \ell$$

The decrease in lengths from unstressed condition are respectively

$$\alpha_1 \ell_1 \Delta T - \Delta \ell \text{ and } \alpha_2 \ell_2 \Delta T + \Delta \ell.$$

The stress developed in the two rods is same. So,

$$Y_1 \left(\frac{\alpha_1 \ell_1 \Delta T - \Delta \ell}{\ell_1} \right) = Y_2 \left(\frac{\alpha_2 \ell_2 \Delta T + \Delta \ell}{\ell_2} \right)$$

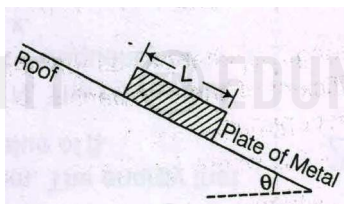
$$\Rightarrow Y_1 \ell_2 (\alpha_1 \ell_1 \Delta T - \Delta \ell) = Y_2 \ell_1 (\alpha_2 \ell_2 \Delta T + \Delta \ell)$$

$$\therefore \Delta \ell = \frac{\ell_1 \ell_2 (Y_1 \alpha_1 - Y_2 \alpha_2) \Delta T}{Y_1 \ell_2 + Y_2 \ell_1}$$

Passage for Question Nons. 9 to 10

A perfectly plane house roof makes an angle θ with the horizontal surface. A flat square metal plate of side L rests over the roof. Each day, temperature varies between day's high (T_1) and day's low (T_2). Coefficients of thermal expansions of roof and plate are α_1 and α_2 ($\alpha_1 < \alpha_2$) respectively. Coefficient of kinetic friction between plate and roof is μ_k .

Plate and roof are always in contact and both have same temperature at every instant of day. Because of difference in coefficients of thermal expansions, each bit of plate is moving relative to the roof, except for points along a certain horizontal line (called stationary line). The stationary line occupies no area, so we assume no force of static friction acting on the plate while the temperature is changing.



9. When the temperature is rising, then distance of the stationary line from top edge of plate will be

$$(a) \frac{L}{2} \left(1 - \frac{\tan \theta}{\mu_k} \right)$$

$$(b) \frac{L}{2} \left(1 + \frac{\tan \theta}{\mu_k} \right)$$

$$(c) \frac{L}{2} \left(\frac{\tan \theta}{\mu_k} \right)$$

$$(d) \frac{L}{2}$$

10. When the temperature is falling, then distance of the stationary line from top edge of plate will be

$$(a) \frac{L}{2} \left(1 - \frac{\tan \theta}{\mu_k} \right)$$

$$(b) \frac{L}{2} \left(1 + \frac{\tan \theta}{\mu_k} \right)$$

$$(c) \frac{L}{2} \left(\frac{\tan \theta}{\mu_k} \right)$$

$$(d) \frac{L}{2}$$

Passage (Solution)

- 9(a) Let the stationary line be at a distance xL from top edge when temperature is rising.

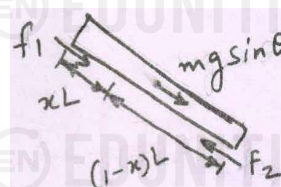
As $\alpha_1 < \alpha_2$, the plate expands more than the roof. So, the friction force f_1 and f_2 (due to upper and lower parts respectively) on the plate are towards the stationary line,

where $f_1 = \mu_k x mg \cos \theta$ and $f_2 = \mu_k (1-x) mg \cos \theta$

As the plate is in equilibrium,

$$mg \sin \theta = f_2 - f_1 = \mu_k mg \cos \theta (1-2x)$$

$$\therefore xL = \frac{L}{2} \left(1 - \frac{\tan \theta}{\mu_k} \right)$$



- 10(b) Let the stationary line be at a distance $x'L$ from top edge when temperature is falling.

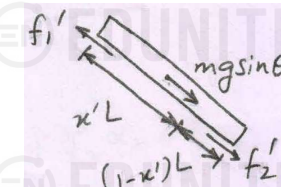
As $\alpha_1 < \alpha_2$, the plate contracts more than the roof. So, the friction forces f'_1 and f'_2 (due to upper and lower parts respectively) on the plate are away from the stationary line where

$f'_1 = \mu_k x' mg \cos \theta$ and $f'_2 = \mu_k (1-x') mg \cos \theta$

As the plate is in equilibrium,

$$mg \sin \theta = f'_1 - f'_2 = \mu_k mg \cos \theta (2x' - 1)$$

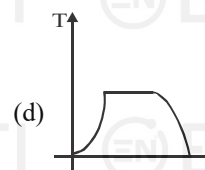
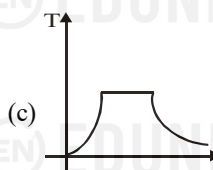
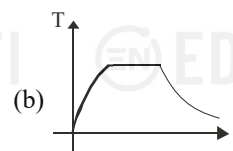
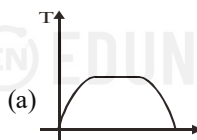
$$\therefore x'L = \frac{L}{2} \left(1 + \frac{\tan \theta}{\mu_k} \right)$$



Passage for Question Nons. 11 to 13

An electric kettle of unknown power rating is switched on after pouring 1 litre water in it. The temperature becomes constant at 60°C after some time. The room temperature is 20°C . When the kettle is switched off, during first 20 s, water cools down by 2°C . Assume Newton's law of cooling to hold. (Specific heat of water is $4200 \text{ J/kg}^\circ\text{C}$)

11. Which is the best graph to represent temperature v/s time?





LEVEL UP - JEE ADVANCED 2022

THERMAL PROPERTIES

EDUNITI

12. What is the wattage of the kettle
(a) 840W (b) 1W
(c) 100W (d) 420 W
13. What is the approximate time taken for the water to cool to 40°C.
(a) 510 s (b) 270 s
(c) 120 s (d) 410 s

Passage (Solution)

- 11(b) After the kettle is switched on, the temperature of water increases and then becomes constant at 60°C. After the kettle is switched off, the temperature falls, initially at a faster rate and finally settles at 20°C.

- 12(d) Let P be the power of the kettle. Before and immediately after the kettle is switched off, the rate of dissipation of heat is same.

$$\Rightarrow P = -ms \frac{dT}{dt} \text{ where } \frac{dT}{dt} = -\frac{2^\circ\text{C}}{20\text{s}} = -0.1^\circ\text{C/s}$$

$$\therefore P = -1 \times 4200 \times 0.1 = 420\text{ W}$$

- 13(b) If T is the instantaneous temperature after the kettle is switched off, then

$$-\frac{dT}{dt} = k(T - 20) \Rightarrow -\int_{60}^T \frac{dT}{T - 20} = \int_0^t k dt$$

$$\Rightarrow t = \frac{1}{k} \ln \left(\frac{60 - 20}{T - 20} \right)$$

$$T = 58^\circ\text{C} \text{ at } t = 2\text{s} \text{ and } T = 40^\circ\text{C} \text{ at } t = t_0$$

$$\Rightarrow 20 = \frac{1}{k} \ln \left(\frac{60 - 20}{58 - 20} \right) = \frac{1}{k} \ln \left(\frac{20}{19} \right) \text{ and}$$

$$t_0 = \frac{1}{k} \ln \left(\frac{60 - 20}{40 - 20} \right) = \frac{1}{k} \ln 2$$

$$\Rightarrow \frac{t_0}{20} = \frac{\ln 2}{\ln(20/19)} = 13.5 \therefore t_0 = 270\text{s}.$$

Passage for Question Nons. 14 to 16

In an experiment of study of Newton's law of cooling, a ball of metal is heated from inside by a heater of 20 W in a room maintained at constant temperature of 20°C.

Following observations are made.

- Mass of the ball is 1 kg.
- Rate of increase of temperature decreases with rise of temperature.
- Temperature of ball becomes 50°C after some time and then remains constant.
- Specific heat of metal is 500 J kg⁻¹ °C⁻¹.

Corresponding to above observations, answer the following questions.

14. Assuming Newton's law of cooling holds for the ball, rate of heat radiated when the ball is at 30°C is
(a) (10/3) W (b) (20/3) W
(c) 10 W (d) (40/3) W

15. The time taken for the temperature to increase from 20°C to 30°C is (Take ln 1.5 = 0.4)
(a) 400 s (b) 300 s
(c) 200 s (d) 600 s
16. During the period, the temperature of ball rises from 20°C to 30°C, total heat energy radiated by the ball is
(a) 2000 J (b) 1500 J
(c) 750 J (d) 1000 J

Passage (Solutions)

- 14(b) Let H be the rate at which heat is radiated. Then,
 $H = k(T - 20)$

$$\text{When } T = 50^\circ\text{C}, 20 = k(50 - 20) \Rightarrow k = (2/3)\text{ W}/^\circ\text{C}$$

$$\text{When } T = 30^\circ\text{C}, H = \frac{2}{3}(30 - 20) = \frac{20}{3}\text{ W}$$

$$15(b) ms \frac{dT}{dt} = 20 - k(T - 20)$$

$$\Rightarrow 1 \times 500 \frac{dT}{dt} = 20 - \frac{2}{3}(T - 20) = \frac{100 - 2T}{3}$$

$$\Rightarrow \int_0^t dt = 750 \int_{20}^{30} \frac{dT}{50 - T} = 750 \ln \left(\frac{50 - 20}{50 - 30} \right)$$

$$\therefore t = 750 \ln 1.5 = 750 \times 0.4 = 300\text{s}.$$

- 16(d) Heat gained = Heat absorbed - Heat radiated

$$\therefore \text{Heat radiated} = \text{Heat absorbed} - \text{Heat gained} \\ = Pt - ms\Delta T = 20 \times 300 - 1 \times 500 \times (30 - 20) \\ = 1000\text{ J}$$

Passage for Question Nons. 17 to 18

A body of area $0.8 \times 10^{-2} \text{ m}^2$ and mass 50g directly faces the sun on a clear day. The body has an emissivity of 0.8 and specific heat of 3000 J/kg K. The surroundings are at 27°C. (solar constant, $G_{sc} = 1.4 \text{ kW/m}^2$, $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$).

17. The rate of rise of the body's temperature (when its temperature is 27°C) is nearly

- (a) 0.36°C/s (b) 0.06°C/s
(c) 0.18°C/s (d) 0.12°C/s

18. The maximum attainable temperature of the body is

- (a) 153°C (b) 396°C
(c) 85°C (d) 40°C

Passage (Solutions)

- 17(b) As per Kirchhoff's law, absorptivity (a) of a body is equal to its emissivity (e). $\Rightarrow a = e = 0.8$

$$ms \frac{dT}{dt} = aG_{sc}A - eA\sigma(T^4 - T_0^4)$$

$$\text{At } T = T_0 = 27^\circ\text{C},$$

$$\frac{dT}{dt} = \frac{aG_{sc}A}{ms} = \frac{0.8 \times 1.4 \times 10^3 \times 0.8 \times 10^{-2}}{0.05 \times 3000} = 0.06^\circ\text{C/s}.$$

- 18(a) At maximum attainable temperature, $dT/dt = 0$

$$\Rightarrow eA\sigma(T^4 - T_0^4) = aG_{sc}A$$

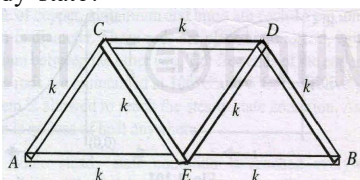
$$\therefore T = \left(\frac{G_{sc}}{\sigma} + T_0^4 \right)^{1/4} = \left(\frac{1.4 \times 10^3}{5.67 \times 10^{-8}} + 300^4 \right)^{1/4} \\ = 426\text{ K} = 153^\circ\text{C}.$$

MORE THAN ONE CORRECT TYPE

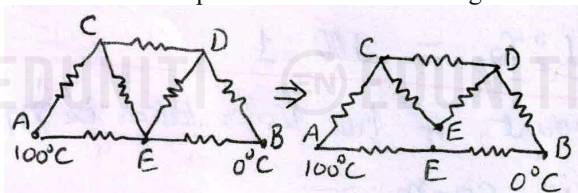
19. Water contained in a jar at room temperature (20°C) is intended to be cooled by method-I or method-II given below
 Method -I: By placing ice cubes and allowing it to float.
 Method-II: By wrapping ice cubes in a wire mesh and allowing it to sink.

Choose best method(s) to cool the water.

- (a) Method-I from 20°C to 4°C
 (b) Method-I from 4°C to 0°C
 (c) Method-II from 20°C to 4°C
 (d) Method-II from 4°C to 0°C
- 19(a, d) Above 4°C , a decrease in temperature increases the density of water. As a result, it descends, replacing the relatively warm water. Convection currents set up in this way demands the location of ice to be on the water surface.
 Below 4°C , a decrease in temperature decreases the water density. As a result, it ascends up displacing the relatively warm water. To setup convection currents in this way, the position of ice cubes should be at the bottom.
20. Seven identical rods of material of thermal conductivity k are connected as shown in fig. All the rods are of identical length l and cross-sectional area A . If one end A is kept at 100°C and the other end B is kept at 0°C , what would be the temperatures (θ_C, θ_D and θ_E) of the junctions C, D and E in the steady state?



- (a) $\theta_C > \theta_E > \theta_D$
 (b) $\theta_E = 50^{\circ}\text{C}$ and $\theta_D = 37.5^{\circ}\text{C}$
 (c) $\theta_E = 50^{\circ}\text{C}, \theta_C = 62.5^{\circ}\text{C}$ and $\theta_D = 37.5^{\circ}\text{C}$
 (d) $\theta_E = 50^{\circ}\text{C}, \theta_C = 60^{\circ}\text{C}$ and $\theta_D = 40^{\circ}\text{C}$
- 20(a, c) Let R be the thermal resistance of each rod. Due to symmetry, the heat currents in rods CE and DC are same. So, we can detach the rods at point E as shown in the figure.



Clearly, $\theta_E = 50^{\circ}\text{C}$ from symmetry.

Now, $R_{AC} = R, R_{CD} = 2R/3$ and $R_{DB} = R$

$$\Rightarrow \frac{100 - \theta_C}{R} = \frac{\theta_C - \theta_D}{2R/3} = \frac{\theta_D - 0}{R} = \frac{100 - 0}{R + 2R/3 + R}$$

$$\therefore \theta_C = 62.5^{\circ}\text{C}, \theta_D = 37.5^{\circ}\text{C} \text{ and } \theta_C > \theta_E > \theta_D$$

21. A hollow copper sphere and a hollow copper cube of same surface area and negligible thickness, are filled with warm water of same temperature and placed in an enclosure of constant temperature, a few degrees below that of the bodies. Then in the beginning,
 (a) the rate of energy lost by the sphere is greater than that by the cube
 (b) the rate of energy lost by the two are equal
 (c) the rate of energy lost by the sphere is less than that by the cube
 (d) the rate of fall of temperature for sphere is less than that for the cube
- 21(b, d) In the beginning, the rate of energy lost (or the power radiated)

$$P = e\sigma A(T^4 - T_0^4)$$

As e, σ, A, T and T_0 is same for both, the rate of energy lost by the two are equal. Also,

$$P = ms \left(-\frac{dT}{dt} \right) \text{ or } -\frac{dT}{dt} = \frac{P}{ms}$$

As the mass of water contained in sphere is more, fall of temperature ($-dT/dt$) for sphere is less than that for the cube.

22. Three bodies A, B and C have equal surface area and thermal emissivities in the ratio $e_A : e_B : e_C = 1 : 1/2 : 1/4$. All the three bodies are radiating at the same rate. Their wavelengths corresponding to maximum intensity are λ_A, λ_B and λ_C respectively and their temperatures are T_A, T_B and T_C on kelvin scale. Then, select the correct statements.

- (a) $\sqrt{T_A T_C} = T_B$
 (b) $\sqrt{\lambda_A \lambda_C} = \lambda_B$
 (c) $\sqrt{e_A T_A} \sqrt{e_C T_C} = e_B T_B$
 (d) $\sqrt{e_A \lambda_A T_A} \sqrt{e_C \lambda_C T_C} = e_B \lambda_B T_B$

- 22(a, b, c, d) Power radiated, $P = e\sigma AT^4$

For same value of P, σ and $A, T \propto (1/e)^{1/4}$

$$\Rightarrow T_A : T_B : T_C = 1 : 2^{1/4} : 4^{1/4}$$

$$\text{Let } T_A = k, T_B = 2^{1/4}k \text{ and } T_C = 4^{1/4}k$$

$$\text{Then, } \sqrt{T_A T_C} = T_B \quad \dots(1)$$

From Wien's displacement law, $\lambda T = \text{const}$

$$\Rightarrow \lambda \propto 1/T$$

$$\Rightarrow \lambda_A : \lambda_B : \lambda_C = 1 : (1/2)^{1/4} : (1/4)^{1/4}$$

$$\text{Let } \lambda_A = c, \lambda_B = (1/2)^{1/4}c \text{ and } \lambda_C = (1/4)^{1/4}c$$

$$\text{Then, } \sqrt{\lambda_A \lambda_C} = \lambda_B \quad \dots(2)$$

$$\sqrt{e_A e_C} = e_B \quad \dots(3)$$

$$\text{From (1) and (3), } \sqrt{e_A T_A} \sqrt{e_C T_C} = e_B T_B$$

$$\text{From (1), (2) and (3), } \sqrt{e_A \lambda_A T_A} \sqrt{e_C \lambda_C T_C} = e_B \lambda_B T_B$$